fn sample_bernoulli_exp1

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn sample_bernoulli_exp1` in `mod.rs` at commit 0be3ab3e6 (up to date). This proof is an adaptation of subsection 5.1 of [CKS20].

1 Vetting history

- Pull Request #519

2 Pseudocode

Precondition

$\mathtt{x} \in \mathbb{Q} \cap (0, 1]$

Implementation

```
def sample_bernoulli_exp1(x) -> bool:
    k = 1
    while True:
        if sample_bernoulli(x / k, false):
            k += 1
        else:
            return is_odd(k)
```

Postcondition

`sample_bernoulli_exp1` returns a boolean sample from $Bernoulli(exp(-x))$, or an error propagated from `sample_bernoulli`.

3 Proof

Define the random variable K^* to be the number of consecutive i, starting from one, for which $B_i \sim Bernoulli(x/i)$ is $\top$.

Lemma 3.1. $P[K^* > n] = \frac{x^n}{n!}$ for any integer $n > 0$ [CKS20].

Proof. Let $A_i \sim Bernoulli(x/i)$ be the sequence of booleans returned from `sample_bernoulli`.

$$
\begin{aligned}
P[K^* > n] &= P[A_1 = A_2 = ... = A_n = \top] && \text{Since } K^* > n, \forall i \leq n, A_i = \top \\
&= \prod_{k=1}^{n} P[A_k = \top] && \text{All } A_i \text{ are independent.} \\
&= \prod_{k=1}^{n} \frac{x}{k} && \text{Since } A_k \sim Bernoulli(x/k) \\
&= \frac{x^n}{n!}
\end{aligned}
$$

□

Theorem 3.2. *is_odd*$(K^*) \sim$ *Bernoulli*$(exp(-x))$ [CKS20].

Proof.

$$
\begin{aligned}
P[K^* \text{ odd}] &= \sum_{k=0}^{\infty} P[K^* = 2k+1] \\
&= \sum_{k=0}^{\infty} (P[K^* > 2k] - P[K^* > 2k+1]) \\
&= \sum_{k=0}^{\infty} \left(\frac{x^{2k}}{(2k)!} - \frac{x^{2k+1}}{(2k+1)!} \right) && \text{by 3.1} \\
&= exp(-x)
\end{aligned}
$$

□

Conditioned on meeting the preconditions, and not returning an error, since `k` is distributed according to K^*, then by 3.2, the return value of `sample_bernoulli_exp1(x)` is distributed *Bernoulli*$(exp(-x))$.

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.